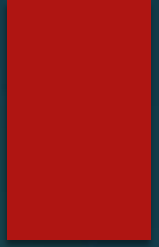




SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)



C01 AT-3

TECHNICAL PRESENTATION

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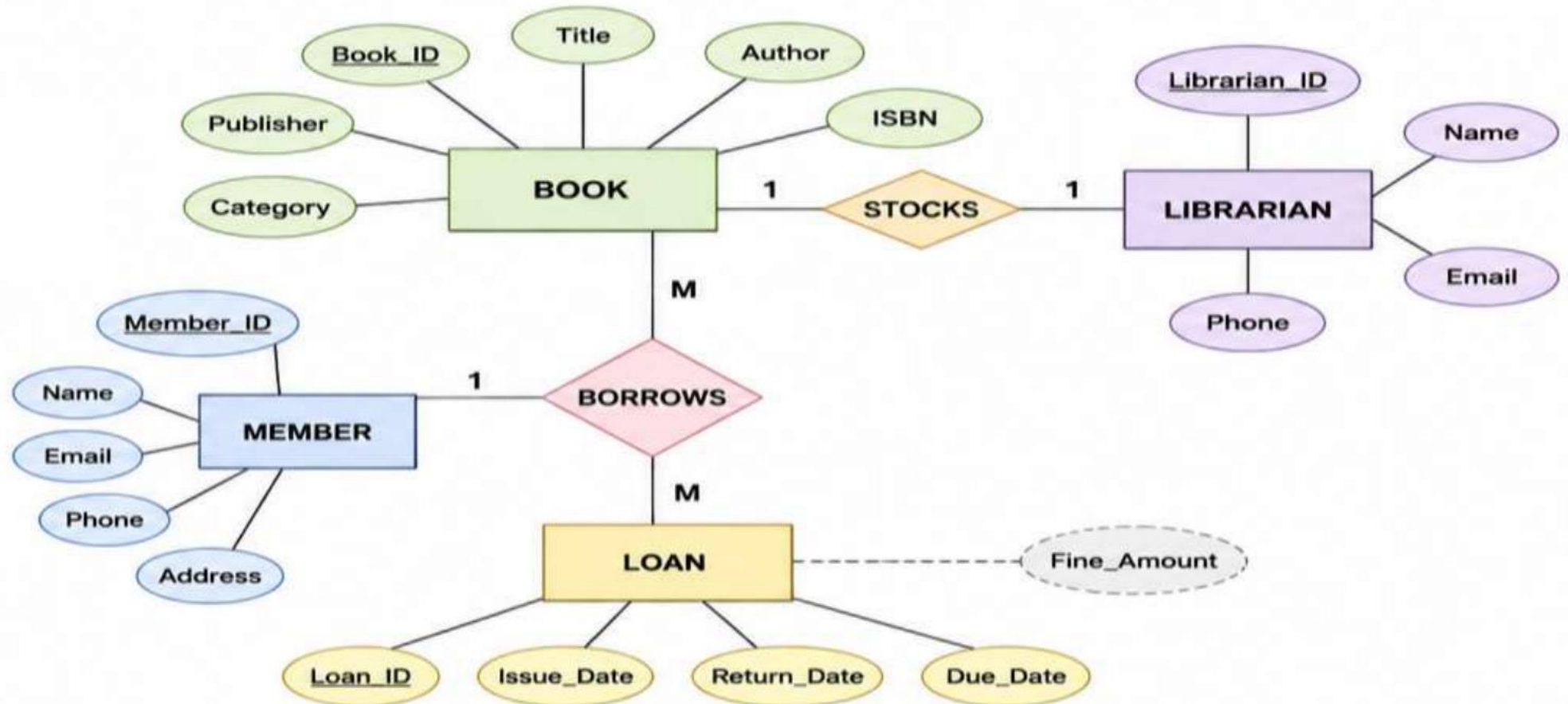


1) Create and present an ER diagram for an Online Library System, explaining each component clearly.

Introduction to Online Library System

- An Online Library System manages books, members, and borrowing activities.
- The ER diagram represents the database structure visually.
- It identifies entities, attributes, and relationships.
- It helps design an efficient and organized database.
- It minimizes data redundancy and improves data consistency.

ER Diagram for an Online Library System



Entity



Relationship



Attribute

1 : One
M : Many

Chen Notation Used

Entities and Attributes

▣ Main Entities

▣ Book

▣ Member

▣ Librarian

▣ Loan

▣ Key Attributes

▣ Book: Book_ID, Title, Author, ISBN

▣ Member: Member_ID, Name, Email

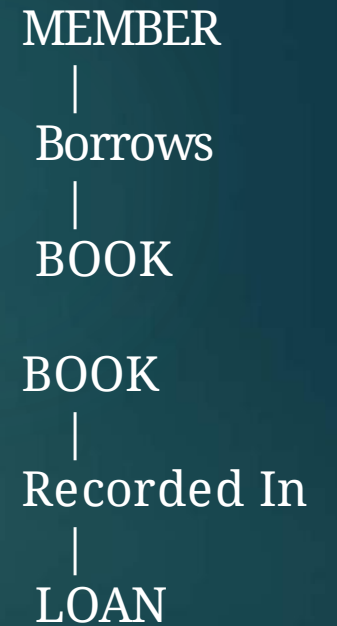
▣ Librarian: Employee_ID, Name

▣ Loan: Loan_ID, Issue_Date, Return_Date

Relationships in ER Diagram

Relationships among Entities

- A Member borrows many Books.
- A Book can be borrowed by different Members over time.
- A Librarian manages book issue and return.
- A Loan records borrowing transactions.
- Relationships define how entities interact.



Advantages and Complete ER Diagram

Benefits of ER Diagram

- ❑ Simplifies database design.
- ❑ Reduces duplicate data.
- ❑ Improves data integrity.
- ❑ Makes future database modifications easier.
- ❑ Supports efficient query processing.





Q2. Prepare a presentation on the EER model, demonstrating generalization and specialization with a University case study.

Introduction to EER Model

Enhanced Entity Relationship (EER) Model

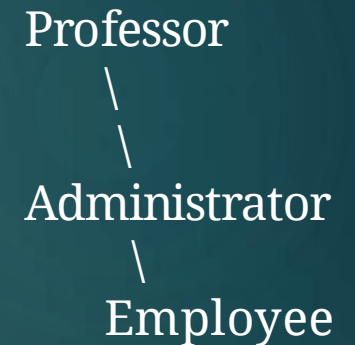
- EER is an extension of the ER model.
- It supports more complex database designs.
- It introduces Generalization and Specialization.
- It models inheritance among entities.
- It represents real-world scenarios more accurately.

Generalization

- ❑ Bottom-up approach.
- ❑ Combines similar entities into one superclass.
- ❑ Common attributes are moved to the parent entity.
- ❑ Reduces redundancy.
- ❑ Improves database organization.

Common Attributes:

- ❑ Employee_ID
- ❑ Name
- ❑ Salary



Specialization

- ❑ Top-down approach.
- ❑ Divides one entity into specialized subclasses.
- ❑ Each subclass has unique attributes.
- ❑ Provides better classification.
- ❑ Supports inheritance.

Example Attributes:

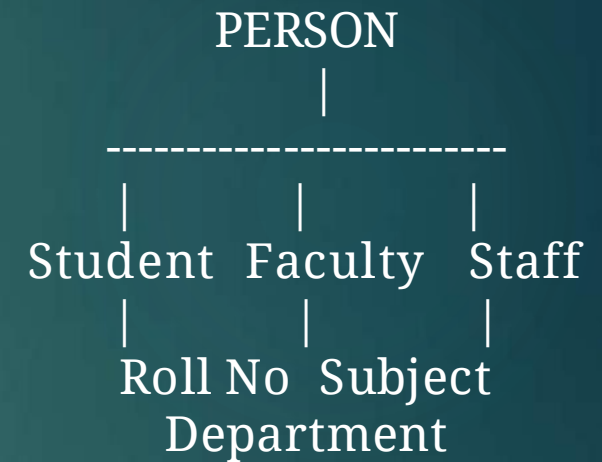
- ❑ Student → CGPA
- ❑ Faculty → Subject
- ❑ Staff → Department



Advantages and University Case Study

Advantages of EER Model

- Represents complex relationships.
- Supports inheritance.
- Reduces data duplication.
- Improves flexibility.
- Suitable for large organizations like universities.



Conclusion

The EER model enhances the traditional ER model by introducing Generalization and Specialization, making university database design more flexible, organized, and efficient.

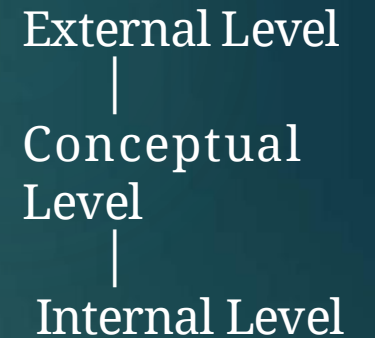


Q3. Present the concept of data independence (logical and physical) and its significance in DBMS architecture.

Introduction to Data Independence

❑ Data Independence in DBMS

- ❑ Data independence is the ability to modify one level of the database without affecting higher levels.
- ❑ It separates logical and physical data storage.
- ❑ It is achieved using the ANSI/SPARC three-level architecture.
- ❑ It improves flexibility and maintainability.
- ❑ It reduces the cost of database modifications.



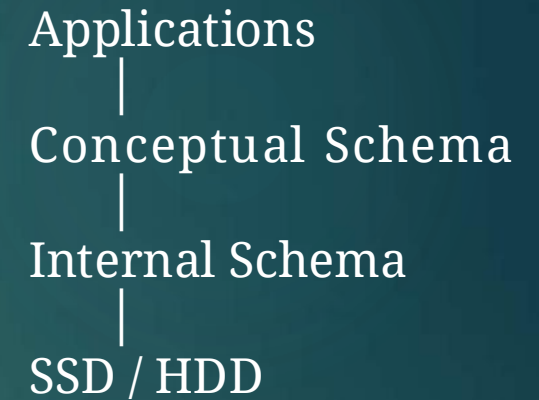
Physical Data Independence

Physical Data Independence

- Physical changes do not affect the conceptual schema.
- Storage devices or file organization can be changed.
- Improves storage efficiency.
- No changes are required in application programs.
- Supported by the Internal Level.

Examples

- Changing HDD to SSD.
- Creating indexes.
- Compressing database files.



Logical Data Independence

- Logical changes do not affect user applications.
- New tables or attributes can be added.
- Existing applications continue working.
- Easier database expansion.
- Supported by the Conceptual Level.

Examples

- Adding Email column.
- Creating a new Student table.
- Adding new relationships.

Student

ID
Name



Student

ID
Name
Email

Significance of Data Independence

Importance of Data Independence

- Reduces maintenance cost.
- Improves database flexibility.
- Supports future database growth.
- Protects application programs from changes.
- Makes database management more efficient.

Physical	Logical
Internal Level	Conceptual Level
Storage changes	Schema changes
Faster implementation	More difficult
No application changes	No user view changes

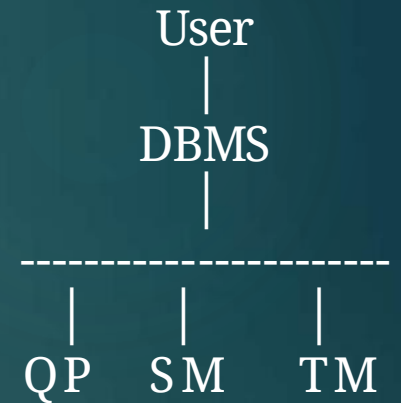


Q4. Deliver a technical presentation on the software components of a DBMS, explaining Query Processor, Storage Manager, and Transaction Manager.

Introduction to DBMS Components

Software Components of DBMS

- DBMS consists of several software modules.
- Each module performs a specific function.
- Major components work together to process data.
- They ensure efficient data management.
- Three important components are Query Processor, Storage Manager, and Transaction Manager.



(QP = Query Processor,
SM = Storage Manager,
TM = Transaction
Manager)

Query Processor

- ☐ Accepts SQL queries from users.
- ☐ Checks syntax and semantics.
- ☐ Optimizes query execution.
- ☐ Generates an execution plan.
- ☐ Retrieves requested data efficiently.

☐ Functions

☐ SQL Parser

☐ Query Optimizer

☐ Execution Engine



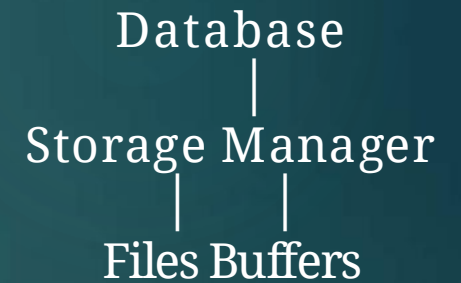
Storage Manager

- Manages physical storage of data.
- Reads and writes data blocks.
- Controls indexes and files.
- Allocates disk space efficiently.
- Ensures secure data storage.

Components

- File Manager
- Buffer Manager
- Authorization Manager

Mini Diagram

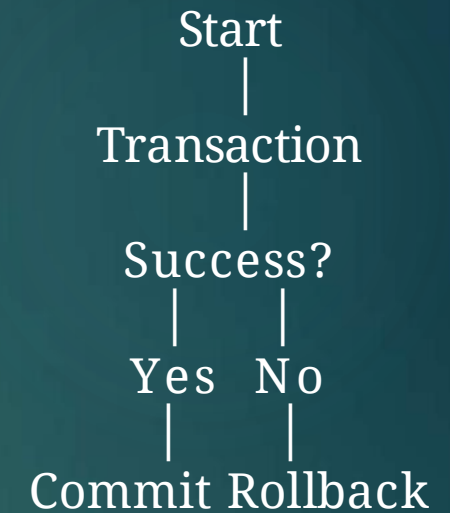


Transaction Manager

- ▣ Controls database transactions.
- ▣ Ensures ACID properties.
- ▣ Handles commit and rollback operations.
- ▣ Maintains data consistency.
- ▣ Supports concurrent user access.

▣ ACID Properties

- ▣ Atomicity
- ▣ Consistency
- ▣ Isolation
- ▣ Durability



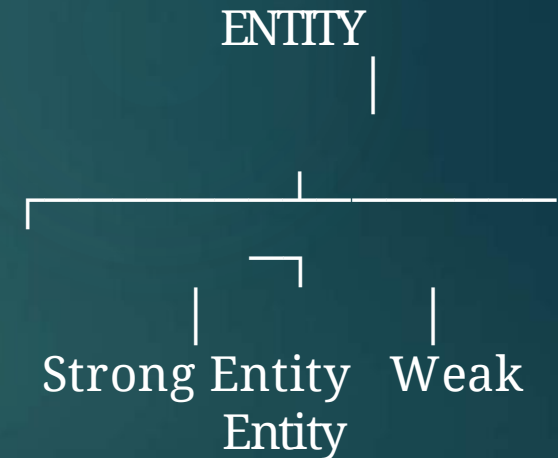


Q5. Prepare a presentation comparing strong and weak entities in ER modeling with real-world examples.

Introduction to Strong and Weak Entities

Strong and Weak Entities in ER Modeling

- ▣ An **Entity** represents a real-world object in a database.
- ▣ Entities are classified into **Strong** and **Weak** entities.
- ▣ Strong entities have their own primary key.
- ▣ Weak entities depend on strong entities for identification.
- ▣ ER diagrams use different symbols to represent them.

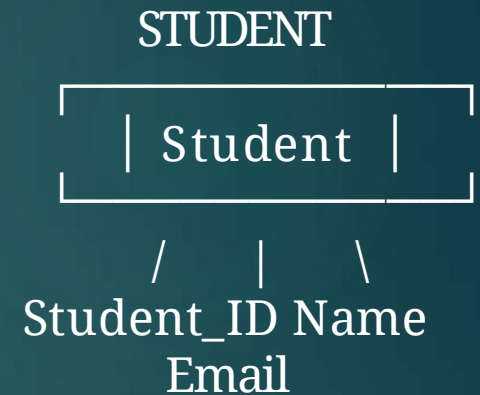


Strong Entity

- Has its own **Primary Key**.
- Exists independently in the database.
- Can be uniquely identified.
- Represented by a **single rectangle** in ER diagrams.
- Used as the parent entity in relationships.

Examples

- Student
- Employee
- Customer
- Product

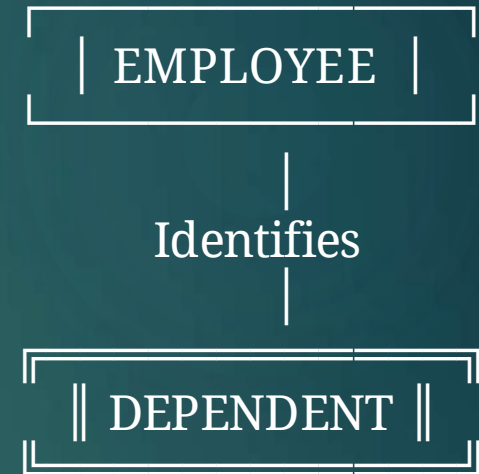


Weak Entity

- ❑ Does **not** have a complete primary key.
- ❑ Depends on a strong entity for identification.
- ❑ Uses a **partial key (discriminator)**.
- ❑ Represented by a **double rectangle**.
- ❑ Connected through an **identifying relationship**.
- ❑ **Real-World Example**
- ❑ Employee → Dependent

Dependent Attributes:

- Dependent_Name
- Age
- Relationship



Comparison of Strong and Weak Entities

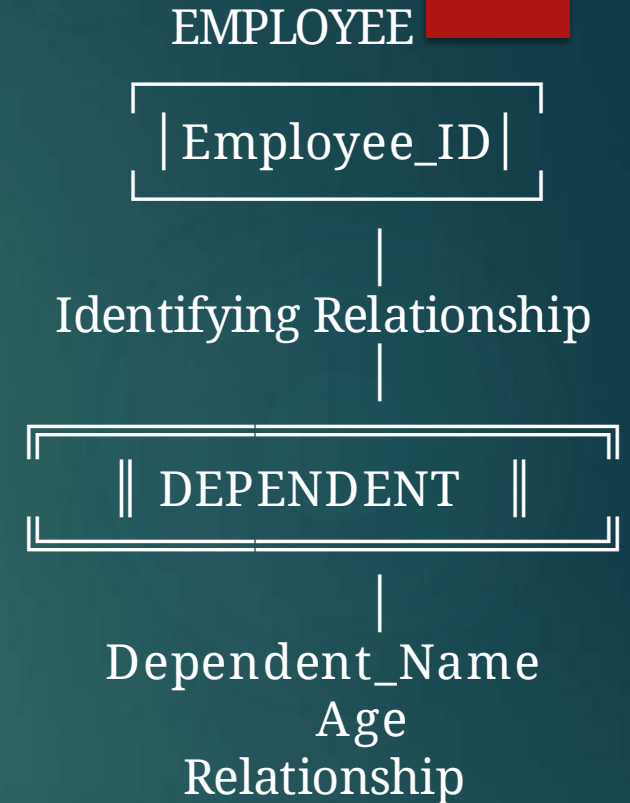
Strong Entity	Weak Entity
Has Primary Key	No Primary Key
Independent	Depends on Strong Entity
Single Rectangle	Double Rectangle
Exists Alone	Cannot Exist Alone
Example: Student	Example: Dependent

Key Points

- Strong entities uniquely identify records.
- Weak entities rely on owner entities.
- Both improve accurate database modeling.
- ER diagrams clearly distinguish both entity types.

Conclusion

Strong entities can exist independently because they have a primary key, whereas weak entities depend on strong entities for identification. Together, they help create accurate and efficient ER models for real-world database systems.





THANK YOU